Supplementary information

Physics-guided learning predicts battery thermal runaway from parsimonious features

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Supplementary Note 1 - Mathematically modeling for thermal runaway

The governing equation is given in Eq. (S1) [1],

$$\sum_{k\in\{p,n\}} m_k c_k T_k + m_{cell} c_{cell} T_{cell} = \dot{Q}_{heater} t + \dot{Q}_{IHG} t - \dot{Q}_{loss} t \tag{1}$$

where m_k denotes the mass of the k-th calorimeter component. Specifically, m_k represents the mass of the copper mesh positioned adjacent to either the positive or negative end of the tested cylindrical cell. c_k and T_k denote the corresponding specific heat capacity and temperature rise of the k-th calorimeter component, respectively. m_{cell}, c_{cell}, and T_{cell} represent the cell mass, specific heat capacity, and temperature rise during the thermal runaway test, respectively. $\dot{Q}_{heater}$, $\dot{Q}_{IHG}$, and $\dot{Q}_{loss}$ denote the heater heat-input rate, internal heat-generation rate of the cell, and heat-loss rate, respectively. Finally, t denotes the elapsed time during the thermal runaway process.

Q_{IHG} denotes the total internal heat generation and is expressed as the sum of the heat contributions from SEI decomposition (Q_{SEI}), anode-electrolyte side reaction (Q_a), electrical internal short-circuit (Q_{ec}), cathode-electrolyte side reaction (Q_c), and electrolyte decomposition (Q_e), as given in Eq. (S2):

$$Q_{IHG} = Q_{SEI} + Q_a + Q_{ec} + Q_c + Q_e \tag{2}$$

Q_{SEI} denotes the heat generated by the exothermic decomposition of the solid electrolyte interphase (SEI), as described in Eq. (S3)

$$\dot{Q}_{SEI} = W_{SEI} H_{SEI} A_{SEI} c_{SEI} \exp\left(-\frac{Ea_{SEI}}{RT}\right) \tag{3}$$

where W_{SEI} is the weighting factor associated with the SEI decomposition reaction, H_{SEI} is the corresponding reaction enthalpy, A_{SEI} is the pre-exponential factor, c_{SEI} is the concentration of the remaining SEI available for decomposition, and Ea_{SEI} is the activation energy of SEI decomposition. R is the universal gas constant $(8.314\ \text{J mol}^{-1}\ \text{K}^{-1})$, and T is the temperature.

Q_a denotes the heat generated by the exothermic reaction between the anode and electrolyte,

$$\dot{Q}_a = W_a H_a A_a c_a \exp\left(-\frac{Ea_a}{RT}\right)\exp\left(-\frac{c_{SEI}}{c_{SEI,ref}}\right) \tag{4}$$

where W_a is the weighting factor associated with the anode active material, H_a is the enthalpy of the anode-electrolyte reaction, A_a is the pre-exponential factor, c_a is the concentration of the remaining unreacted anode material, and Ea_a is the activation energy of the anode-electrolyte reaction. c_{SEI} denotes the concentration of the remaining SEI available for the anode-electrolyte reaction. R is the universal gas constant $(8.314\ \text{J mol}^{-1}\ \text{K}^{-1})$, and T is the temperature.

Q_{ec} denotes the heat generated by internal electrical short-circuiting, as described in Eq. (S5) [2]:

$$\dot{Q}_{ec} = \frac{3600\eta VC}{V_{cell}} T_{ISC} A_{ec} SOC \exp\left(-\frac{Ea_{ec}}{RT}\right) \tag{5}$$

where η is the internal short-circuit (ISC) factor, T_{ISC} is the ISC triggering temperature, A_{ec} is the pre-exponential factor, and Ea_{ec} is the activation energy associated with the ISC process V, C, V_{cell}, and SOC denote the open-circuit voltage, cell capacity, cell volume, and state of

charge, respectively, all of which can be obtained prior to thermal runaway testing. R is the universal gas constant $(8.314\ \text{J mol}^{-1}\ \text{K}^{-1})$, and T is the temperature.

Q_c denotes the heat generated by the exothermic reaction between the delithiated cathode material and the electrolyte, as given in Eq. (S6)

$$\dot{Q}_c = W_c H_c A_c \alpha_c \left(1-\alpha_c\right) \exp\left(-\frac{Ea_c}{RT}\right) \tag{6}$$

where W_c is the weighting factor associated with the cathode active material, H_c is the enthalpy of the cathode-electrolyte reaction, and A_c is the pre-exponential factor. α_c represents the dimensionless reacted fraction of the delithiated cathode material, whereas $1-\alpha_c$ represents the corresponding unreacted fraction. Ea_c is the activation energy of the cathode-electrolyte reaction, R is the universal gas constant $(8.314\ \text{J mol}^{-1}\ \text{K}^{-1})$, and T is the temperature.

Q_e denotes the heat generated by the exothermic decomposition of the electrolyte, as given in Eq. (S7)

$$\dot{Q}_e = W_e H_e A_e c_e \exp\left(-\frac{Ea_e}{RT}\right) \tag{7}$$

where W_e is the weighting factor associated with the electrolyte, H_e is the enthalpy of electrolyte decomposition, A_e is the pre-exponential factor, c_e is the concentration of the remaining undecomposed electrolyte, and Ea_e is the activation energy of electrolyte decomposition. R is the universal gas constant $(8.314\ \text{J mol}^{-1}\ \text{K}^{-1})$, and T is the temperature.

$\dot{Q}_{loss}$ denotes the total heat-loss rate and is expressed as the sum of the convective, radiative, and conductive heat-loss rates, as given in Eq. (S8).

$$\dot{Q}_{loss} = \dot{Q}_{conv} + \dot{Q}_{rad} + \dot{Q}_{cond} \tag{8}$$

$\dot{Q}_{conv}$ denotes the convective heat-loss rate, as described in Eq. (S9):

$$\dot{Q}_{conv} = hA\left(T_S - T_\infty\right) \tag{9}$$

where h is the convective heat-transfer coefficient, A is the surface area of the cylindrical cell, T_s is the cell-surface temperature, and T_∞ is the ambient temperature of the surrounding fluid sufficiently far from the cell surface.

$\dot{Q}_{rad}$ denotes the radiative heat-loss rate, as given in Eq. (S10)

$$\dot{Q}_{rad} = \varepsilon\sigma A\left(T_S^4 - T_\infty^4\right) \tag{10}$$

where ε is the surface emissivity, σ is the Stefan-Boltzmann constant, A is the radiating surface area, T_s is the absolute surface temperature of the cell, and T_∞ is the absolute temperature of the surroundings.

$\dot{Q}_{cond}$ denotes the conductive heat-loss rate, as described in Eq. (S11)

$$\dot{Q}_{cond} = \sum_i \left(\frac{k_i A_i}{L_i}\right)\left(T_S - T_\infty\right) \tag{11}$$

where k_i is the thermal conductivity of the material along the i-th heat-transfer path, A_i is the corresponding cross-sectional area, and L_i is the effective conduction length. T_s and T_∞ denote the cell-surface temperature and ambient temperature, respectively.

Figures

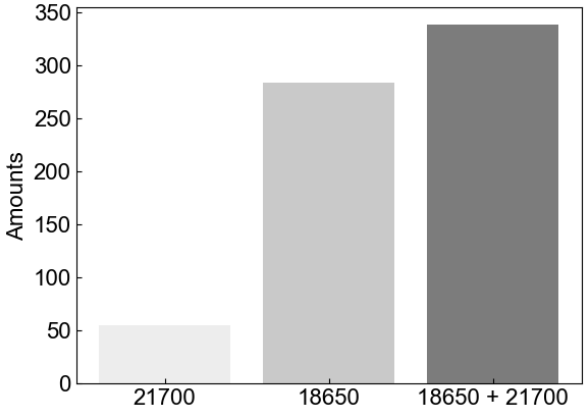


Fig. S1 Cell information [3].

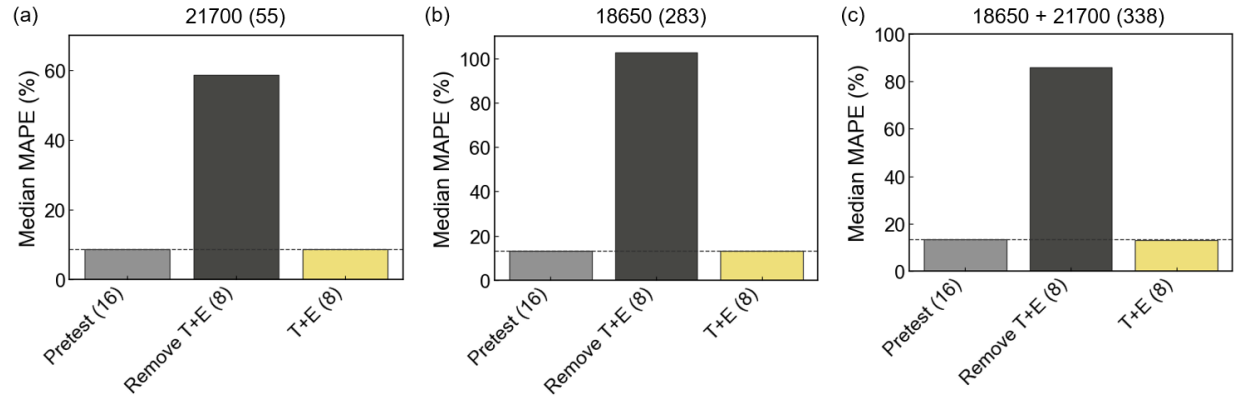


Fig. S2 Comparison between with physical features and without physical features. (a) Prediction results of 21700 cells. (b) Prediction results of 18650 cells. (c) Prediction results of 18650 + 21700 cells. T and E denote thermal and electrochemical features, respectively.

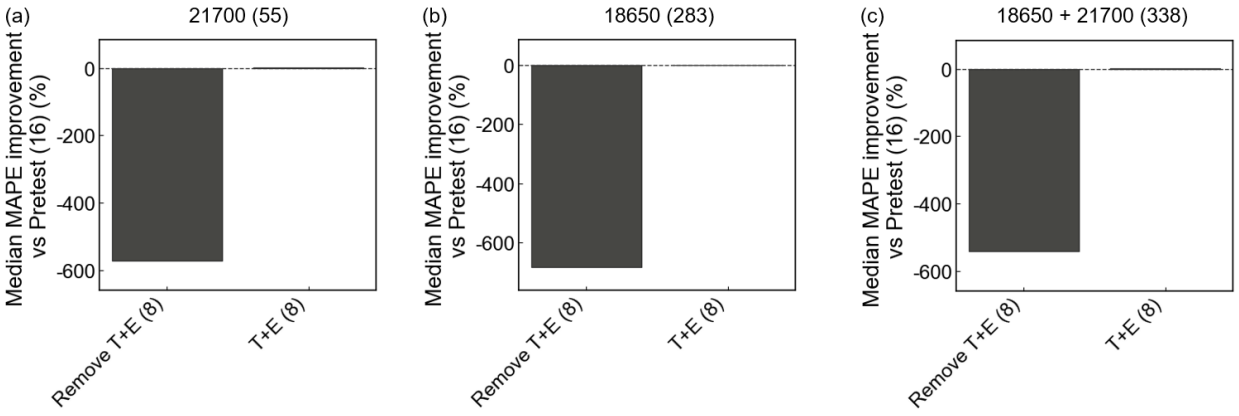


Fig. S3 Comparison of prediction improvement ratio between with physical features and without physical features. (a) Prediction results of 21700 cells. (b) Prediction results of 18650 cells. (c) Prediction results of 18650 + 21700 cells. T and E denote thermal and electrochemical features, respectively.

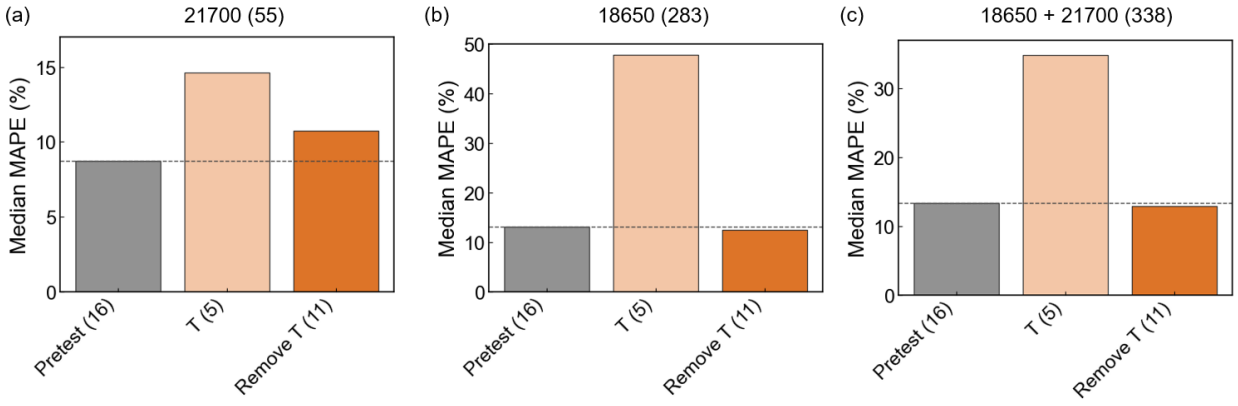


Fig. S4 Comparison between with thermal features and without thermal features. (a) Prediction results of 21700 cells. (b) Prediction results of 18650 cells. (c) Prediction results of 18650 + 21700 cells. T and E denote thermal and electrochemical features, respectively.

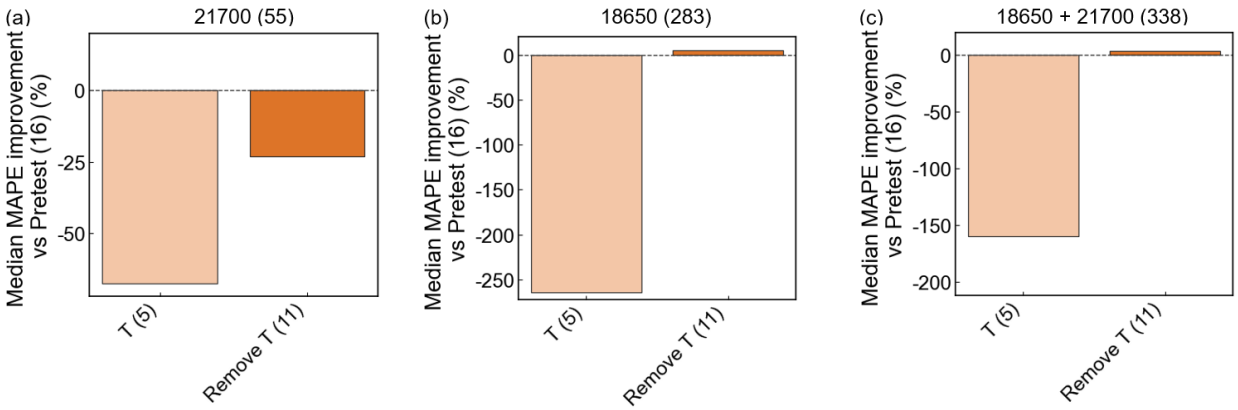


Fig. S5 Comparison of prediction improvement ratio between with thermal features and without thermal features. (a) Prediction results of 21700 cells. (b) Prediction results of 18650 cells. (c) Prediction results of 18650 + 21700 cells. T and E denote thermal and electrochemical features, respectively.

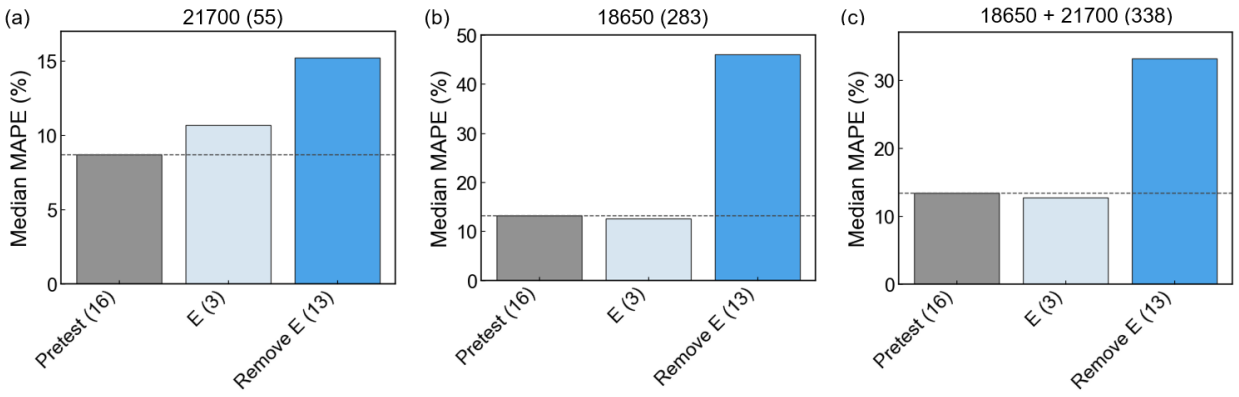


Fig. S6 Comparison between with electrochemical features and without electrochemical features. (a) Prediction results of 21700 cells. (b) Prediction results of 18650 cells. (c) Prediction results of 18650 + 21700 cells. T and E denote thermal and electrochemical features, respectively.

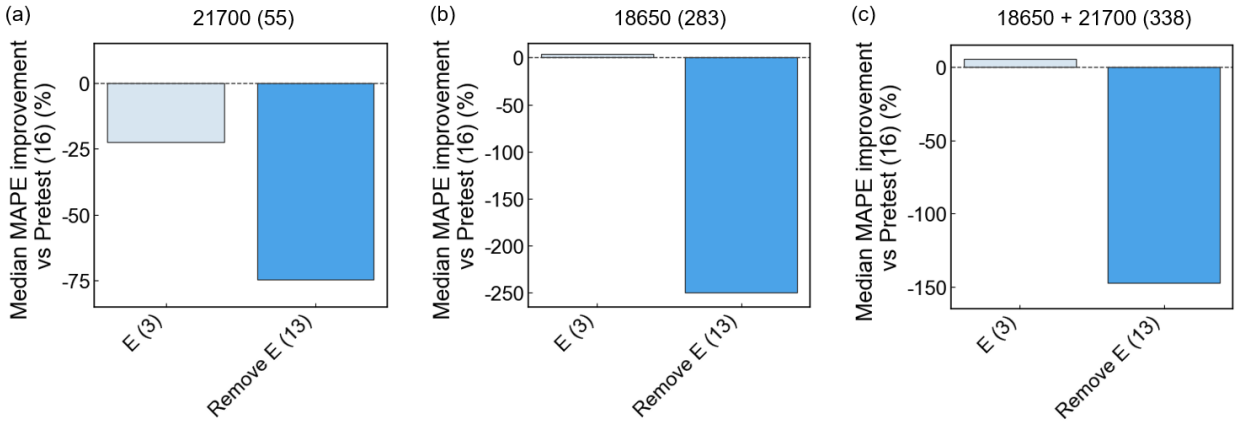


Fig. S7 Comparison of prediction improvement ratio between with electrochemical features and without electrochemical features. (a) Prediction results of 21700 cells. (b) Prediction results of 18650 cells. (c) Prediction results of 18650 + 21700 cells. T and E denote thermal and electrochemical features, respectively.

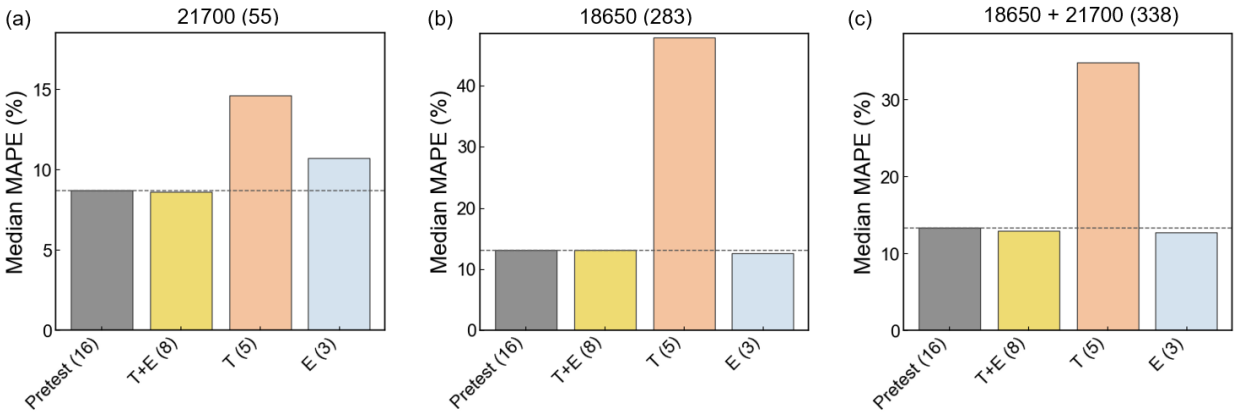


Fig. S8 Comparison among with different physical features. (a) Prediction results of 21700 cells. (b) Prediction results of 18650 cells. (c) Prediction results of 18650 + 21700 cells. T and E denote thermal and electrochemical features, respectively.

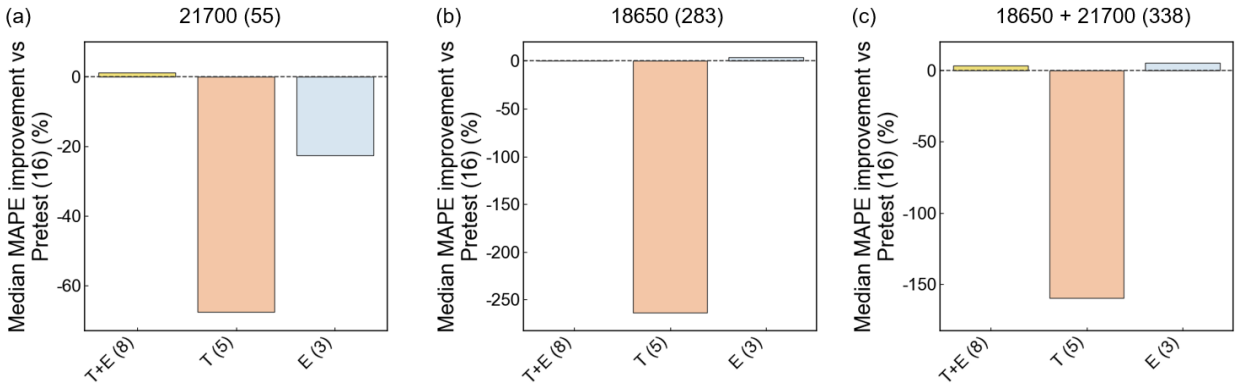


Fig. S9 Comparison of prediction improvement ratio among different physical features. (a) Prediction results of 21700 cells. (b) Prediction results of 18650 cells. (c) Prediction results of 18650 + 21700 cells. T and E denote thermal and electrochemical features, respectively.

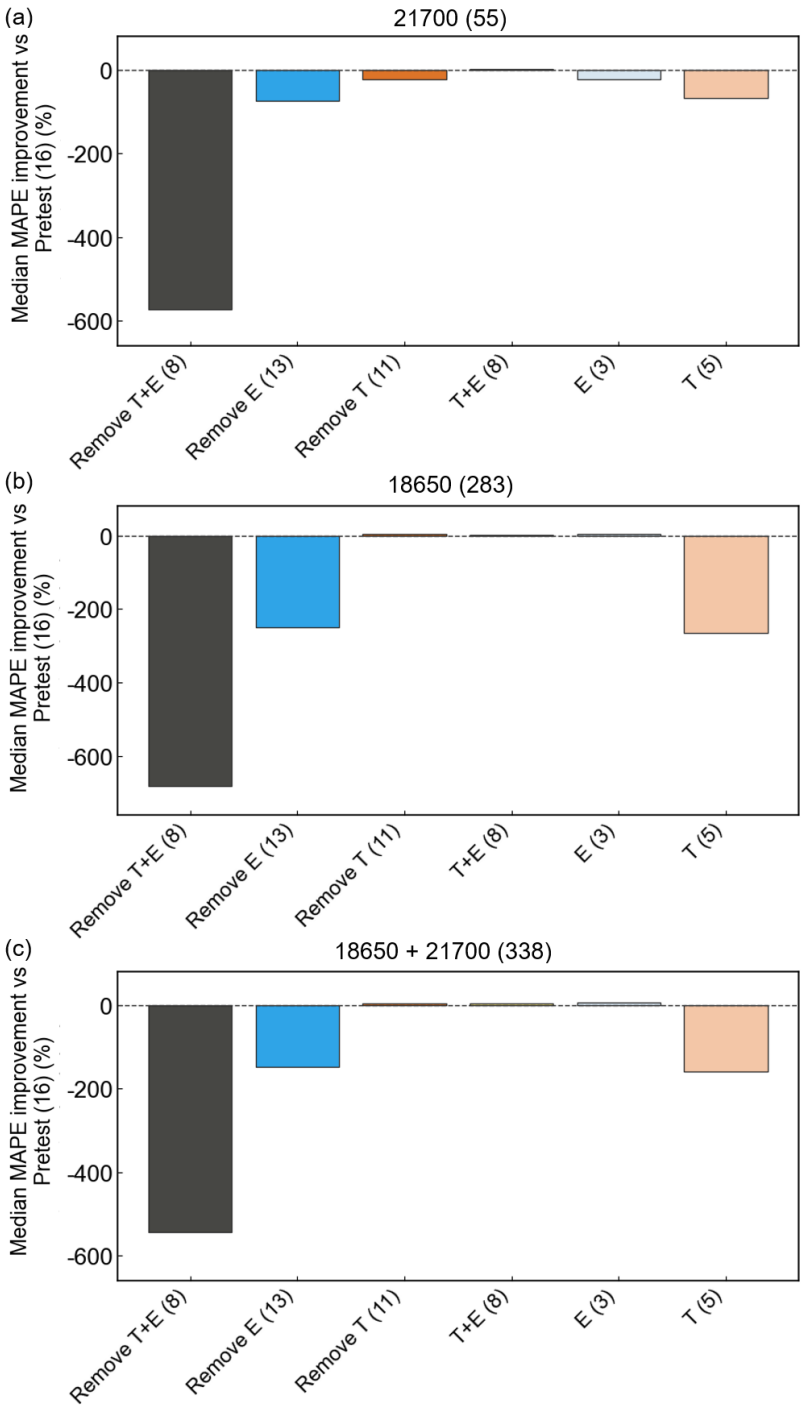


Fig. S10 Comparison of prediction improvement ratio among different physical features. (a) Prediction results of 21700 cells. (b) Prediction results of 18650 cells. (c) Prediction results of 18650 + 21700 cells. T and E denote thermal and electrochemical features, respectively.

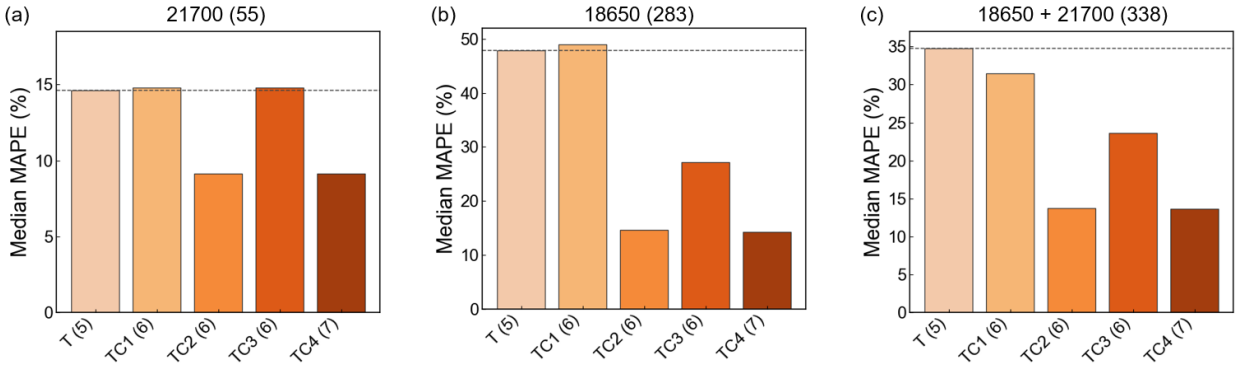


Fig. S11 Comparison among thermal features with different constructed features. (a) Prediction results of 21700 cells. (b) Prediction results of 18650 cells. (c) Prediction results of 18650 + 21700 cells.

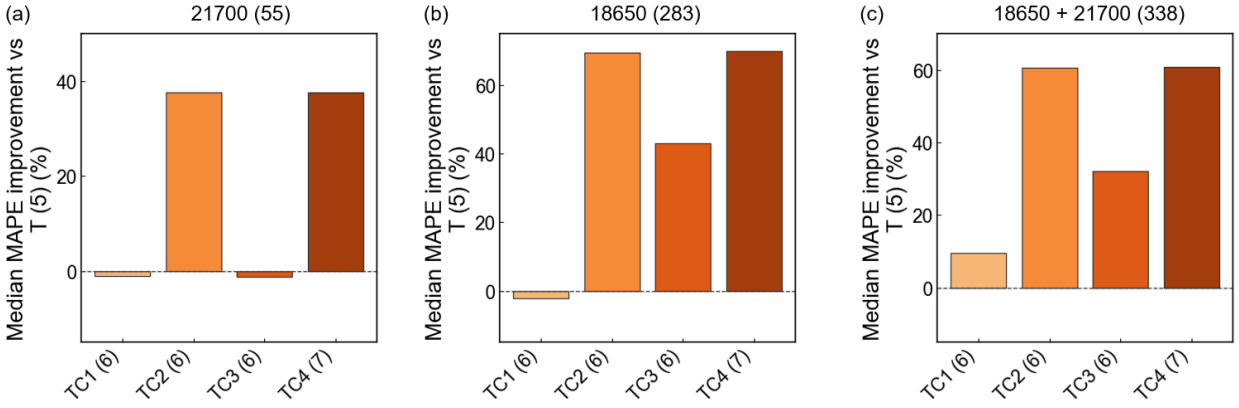


Fig. S12 Comparison of prediction improvement ratio among thermal features with different constructed features. (a) Prediction results of 21700 cells. (b) Prediction results of 18650 cells. (c) Prediction results of 18650 + 21700 cells.

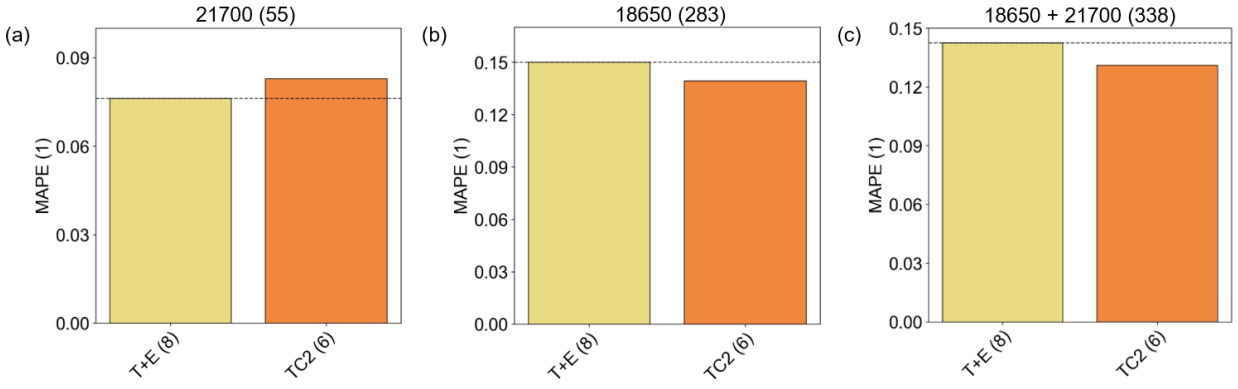


Fig. S13 Comparison of feature space with thermal and electrochemical information between purely selected features and selected + constructed features. (a) Prediction results of 21700 cells. (b) Prediction results of 18650 cells. (c) Prediction results of 18650 + 21700 cells.

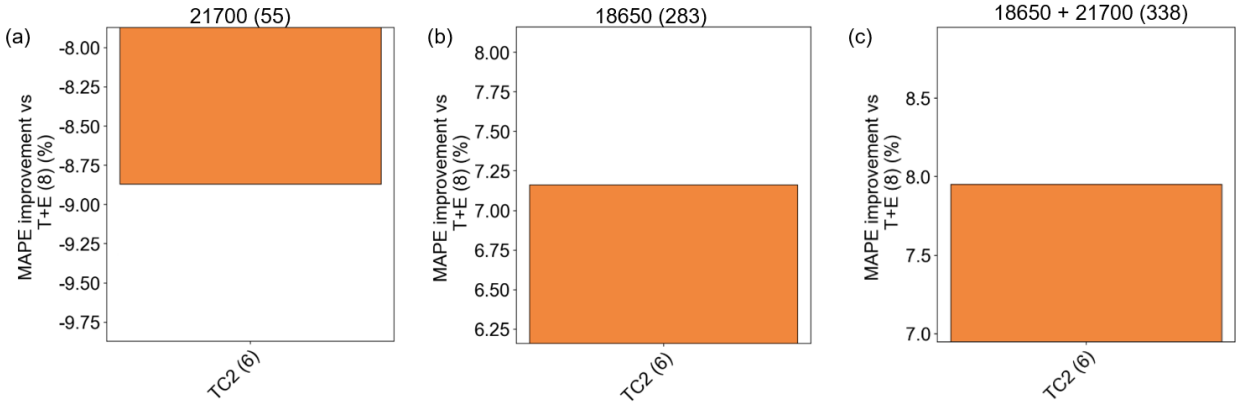


Fig. S14 Comparison of prediction improvement ratio of selected + constructed features over purely selected features. (a) Prediction results of 21700 cells. (b) Prediction results of 18650 cells. (c) Prediction results of 18650 + 21700 cells.

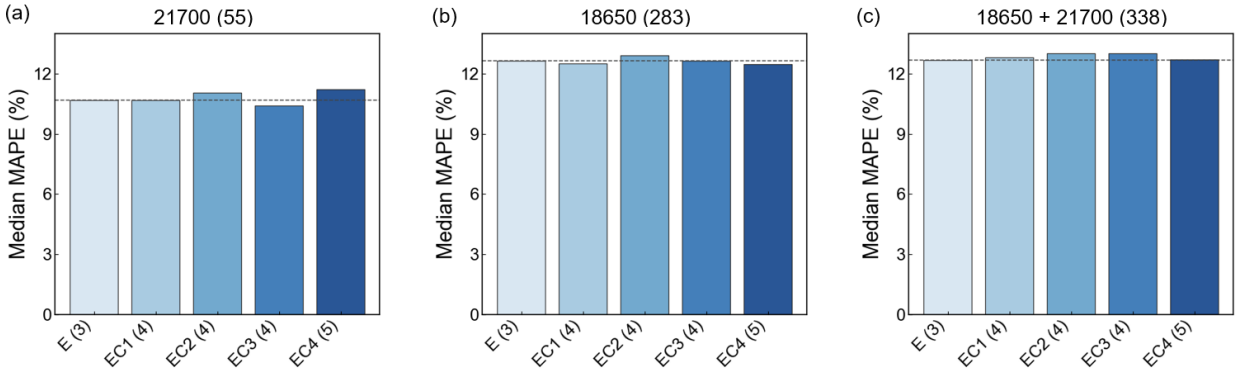


Fig. S15 Comparison among electrochemical features with different constructed features. (a) Prediction results of 21700 cells. (b) Prediction results of 18650 cells. (c) Prediction results of 18650 + 21700 cells.

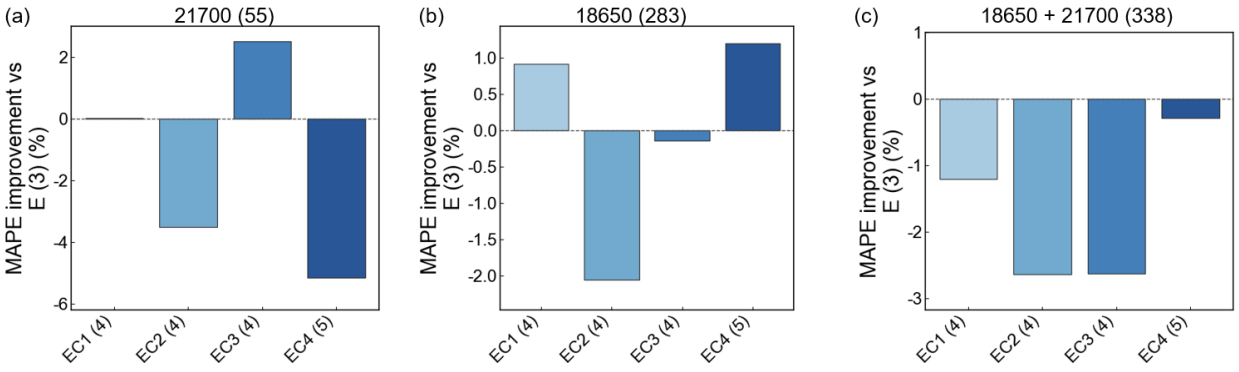


Fig. S16 Comparison of prediction improvement ratio among electrochemical features with different constructed features. (a) Prediction results of 21700 cells. (b) Prediction results of 18650 cells. (c) Prediction results of 18650 + 21700 cells.

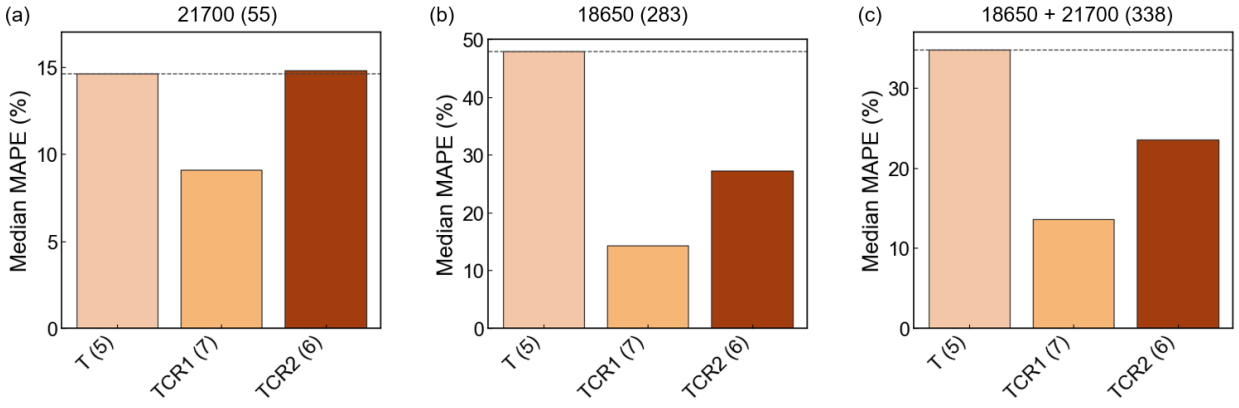


Fig. S17 Comparison among thermal features with different constructed features (without redundancy). (a) Prediction results of 21700 cells. (b) Prediction results of 18650 cells. (c) Prediction results of 18650 + 21700 cells.

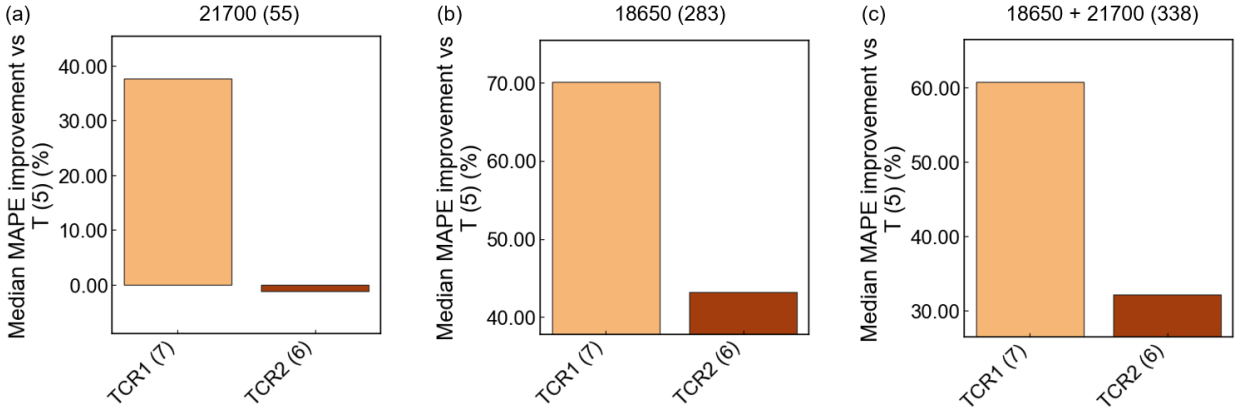


Fig. S18 Comparison of prediction improvement ratio among thermal features with different constructed features (without redundancy). (a) Prediction results of 21700 cells. (b) Prediction results of 18650 cells. (c) Prediction results of 18650 + 21700 cells.

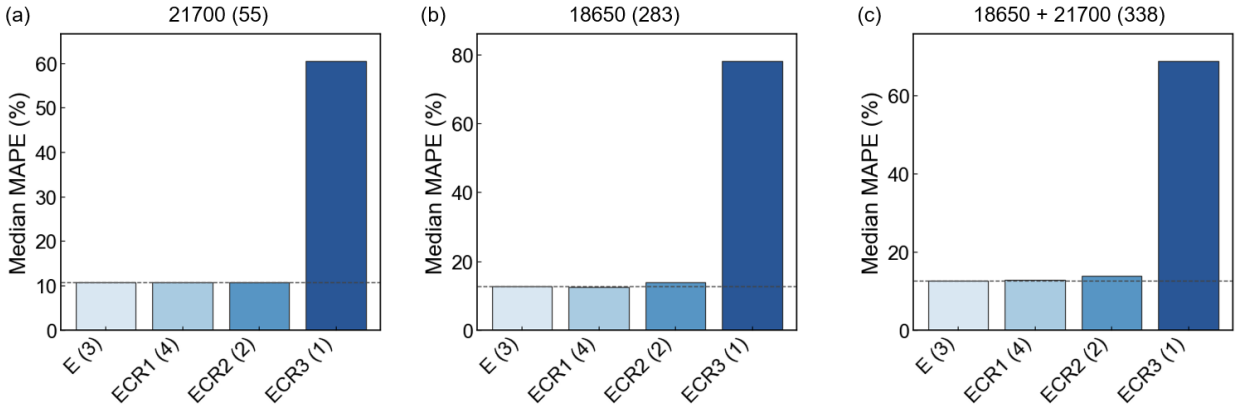


Fig. S19 Comparison among electrochemical features with different constructed features (without redundancy). (a) Prediction results of 21700 cells. (b) Prediction results of 18650 cells. (c) Prediction results of 18650 + 21700 cells.

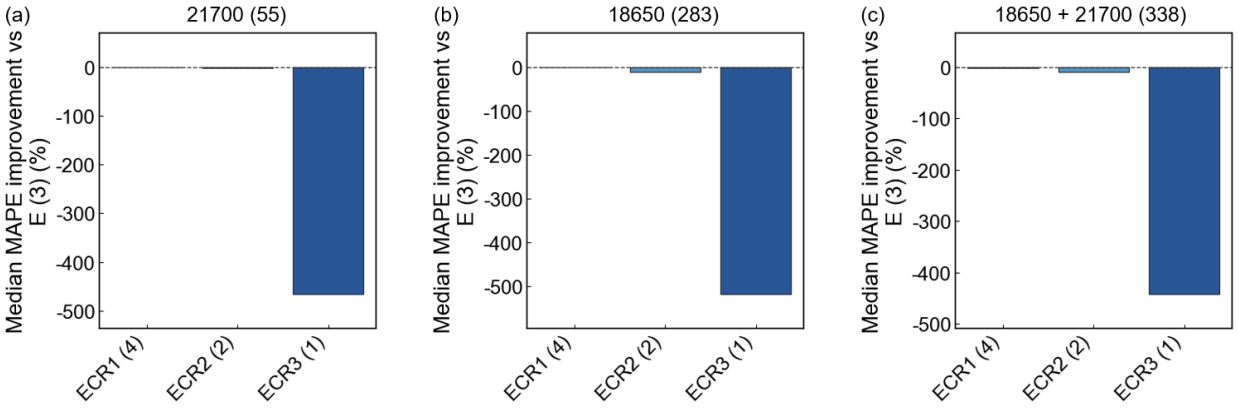


Fig. S20 Comparison of prediction improvement ratio among electrochemical features with different constructed features (without redundancy). (a) Prediction results of 21700 cells. (b) Prediction results of 18650 cells. (c) Prediction results of 18650 + 21700 cells.

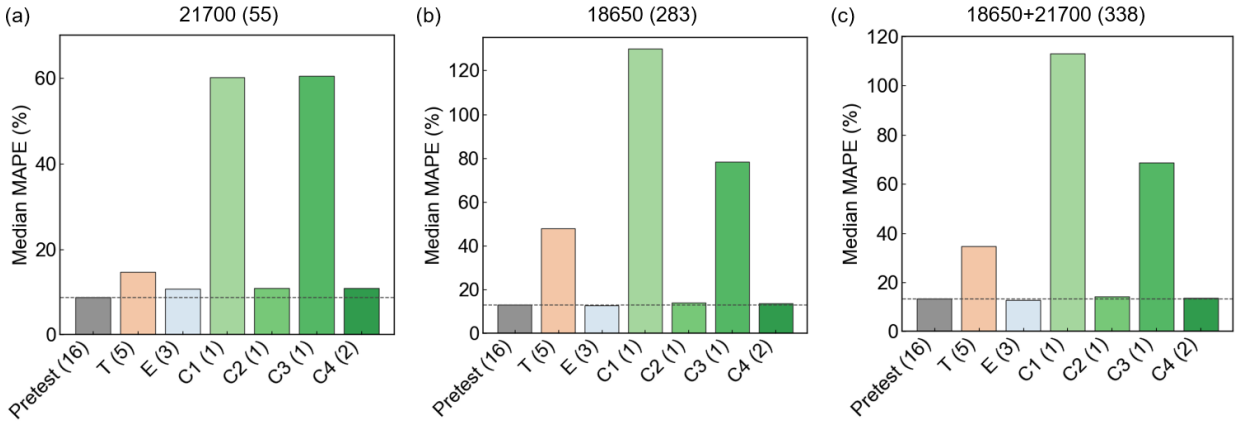


Fig. S21 Comparison among different constructed features. (a) Prediction results of 21700 cells. (b) Prediction results of 18650 cells. (c) Prediction results of 18650 + 21700 cells.

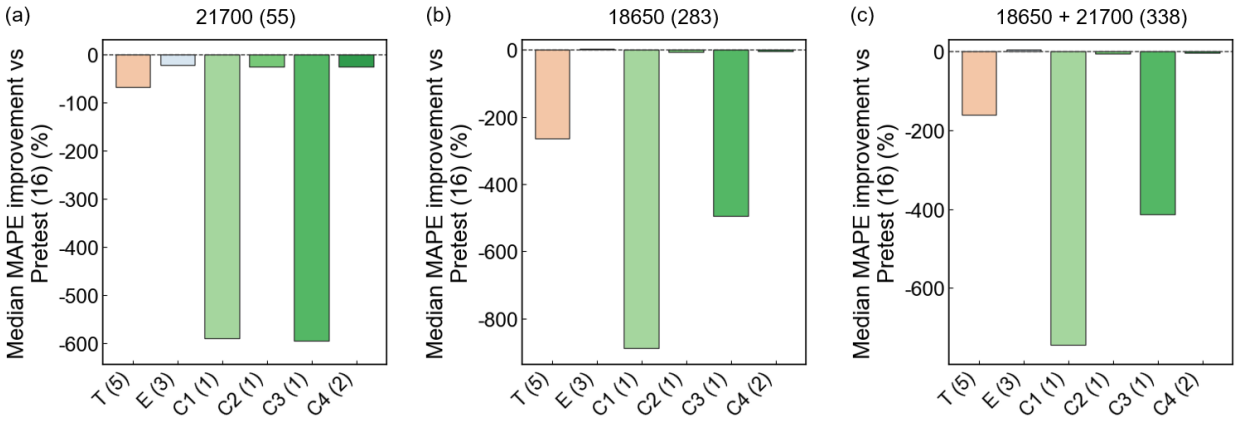


Fig. S22 Comparison of prediction improvement ratio among different constructed features. (a) Prediction results of 21700 cells. (b) Prediction results of 18650 cells. (c) Prediction results of 18650 + 21700 cells.

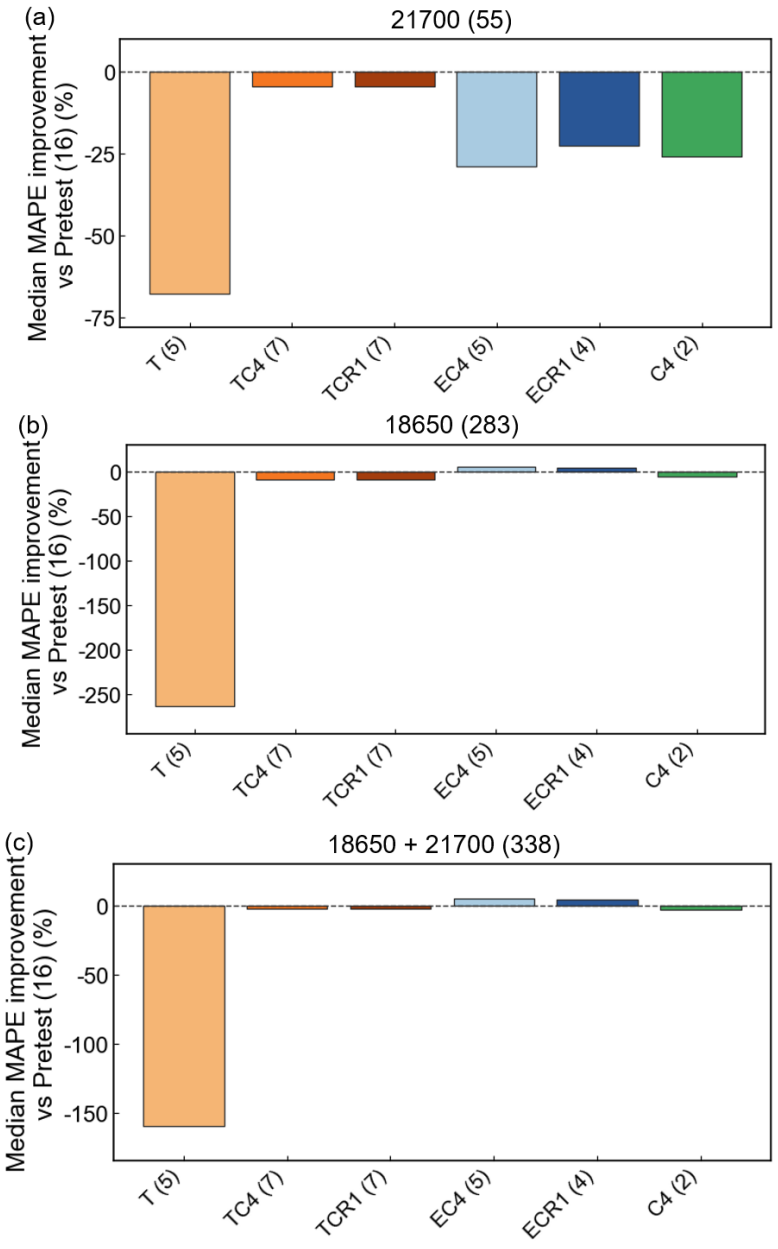


Fig. S23 Summary of best prediction improvement ratio among different selected feature and constructed features combination. (a) Prediction results of 21700 cells. (b) Prediction results of 18650 cells. (c) Prediction results of 18650 + 21700 cells. T and E denote thermal and electrochemical features, respectively.

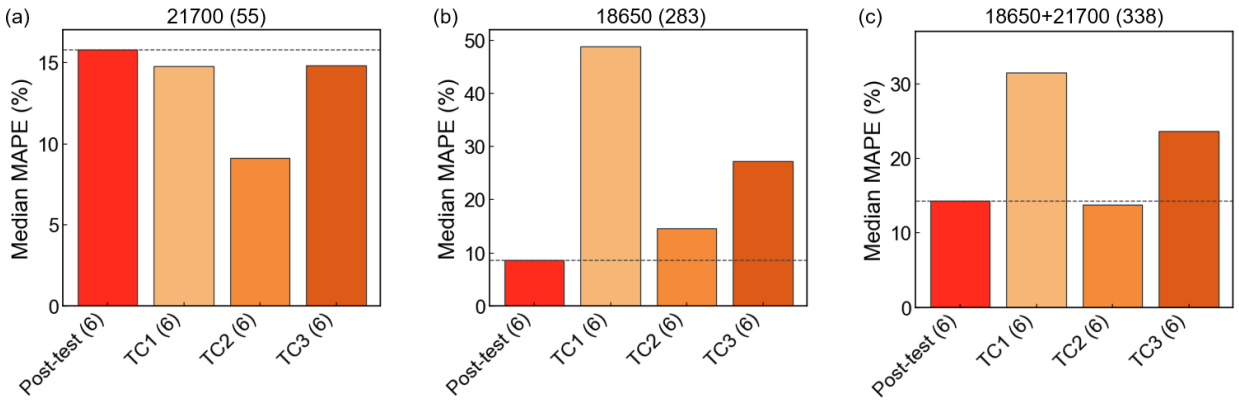


Fig. S24 Comparison of 6 features with post-test features and pretest features (TC1, TC2, and TC3). (a) Prediction results of 21700 cells. (b) Prediction results of 18650 cells. (c) Prediction results of 18650 + 21700 cells.

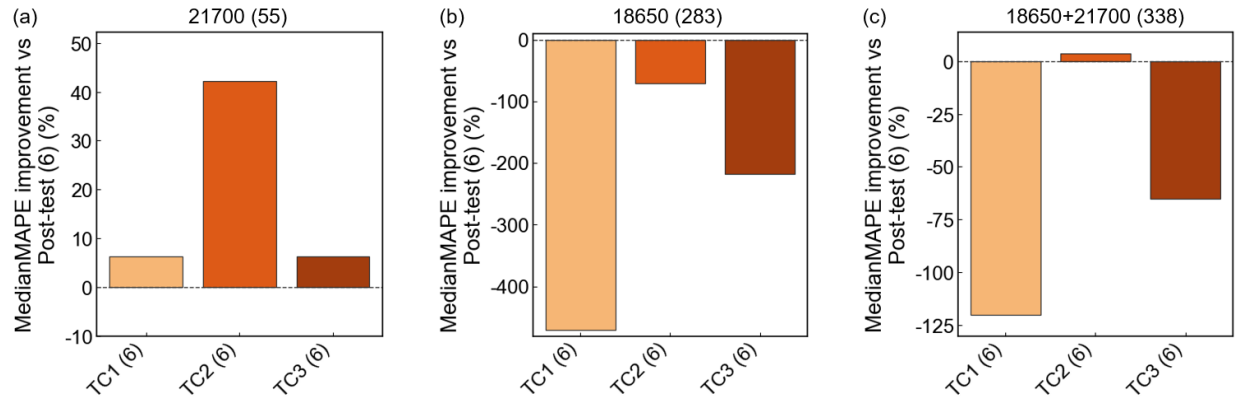


Fig. S25 Comparison of 6 features with post-test features and pretest features (TC1, TC2, and TC3). (a) Prediction results of 21700 cells. (b) Prediction results of 18650 cells. (c) Prediction results of 18650 + 21700 cells.

Tables

Table. S1 Cell description [3]

No.	Cell-Description	Count
1	LG 18650-MJ1 (Korean)	35
2	MOLiCEL 18650-P28A	24
3	KULR 18650-K330	22
4	LG 18650-M36	19
5	LG 18650-Test Cell (BV-220)	16
6	LG 18650-Test Cell (BV-250)	14
7	Sanyo 18650-A	12
8	Sony 18650-VC7	12
9	Soteria 18650 (Control)	12
10	LG 18650-Test Cell (NBV-250)	11
11	Soteria 18650 (CU)	11
12	MOLiCEL 18650-M35A	10
13	Samsung 18650-26J	10
14	Samsung 18650-30Q	10
15	Soteria 18650 (AL)	9
16	Soteria 18650 (ALCUDW)	9
17	Soteria 18650 (DW)	9
18	LG 18650-Test Cell (NBV-220)	6
19	Soteria 18650 (ALCU)	6

20	Soteria 18650 (ALDW)	5
21	LG 18650-HG2	4
22	MOLiCEL 18650-J	4
23	Sony 18650-VTC6	4
24	MOLiCEL 18650-Test Cell (DW-Gold)	3
25	MOLiCEL 18650-Test Cell (DW-Silver)	3
26	Panasonic 18650-BE	2
27	MOLiCEL 18650-Test Cell	1
28	KULR 21700-K500	30
29	LG 21700-M50 (BV)	25

Table. S2 Feature selection test design

Selected feature	Definition	Unit	Source term
OCV	Pretest open-circuit voltage	V	$\dot{Q}_{IHG}$
Q	Pretest capacity	Ah	$\dot{Q}_{IHG}$
SOC	Pretest state of charge	%	$\dot{Q}_{IHG}$
m_{cell}	Pretest cell mass	g	m_{cell}
m_p	Positive copper mesh mass	g	m_p
m_n	Negative copper mesh mass	g	m_n
Q_{heater}	Heater power	W	$\dot{Q}_{heater}$
t_{heater}	Heater time	s	t
E	Energy	Wh	$\dot{Q}_{IHG}$
$\rho_{E,V}$	Energy density	$\frac{Wh}{L}$	$\dot{Q}_{IHG}$
A_{side}	Surface area casing side	cm^2	$\dot{Q}_{loss}$
A_{top}	Surface area top	cm^2	$\dot{Q}_{loss}$
A_{bottom}	Surface area bottom	cm^2	$\dot{Q}_{loss}$
A_{total}	Surface area total	cm^2	$\dot{Q}_{loss}$
$V_{nominal}$	Nominal voltage	V	
$\rho_{E,m}$	Specific energy	$\frac{Wh}{kg}$	

Table. S3 Feature selection test design

Test Name	Features	Definition
E	OCV	Pretest open-circuit voltage
	Q	Pretest capacity
	SOC	Pretest state of charge
T	m_{cell}	Pretest cell mass
	$m_{cm,p}$	Positive copper mesh mass
	$m_{cm,n}$	Negative copper mesh mass
	Q_{heater}	Heater power
	t_{heater}	Heater time
T+E	m_{cell}	Pretest cell mass
	$m_{cm,p}$	Positive copper mesh mass
	$m_{cm,n}$	Negative copper mesh mass
	Q_{heater}	Heater power
	t_{heater}	Heater time
	OCV	Pretest open-circuit voltage
	Q	Pretest capacity
	SOC	Pretest state of charge
Remove E	m_{cell}	Pretest cell mass
	$m_{cm,p}$	Positive copper mesh mass
	$m_{cm,n}$	Negative copper mesh mass

	Q_{heater}	Heater power
	t_{heater}	Heater time
	E	Energy
	$\rho_{E,V}$	Energy density
	A_{side}	Surface area casing side
	A_{top}	Surface area top
	A_{bottom}	Surface area bottom
	A_{total}	Surface area total
	E	Energy
	$\rho_{E,V}$	Energy density
Remove T	OCV	Pretest open-circuit voltage
	Q	Pretest capacity
	SOC	Pretest state of charge
	E	Energy
	$\rho_{E,V}$	Energy density
	A_{side}	Surface area casing side
	A_{top}	Surface area top
	A_{bottom}	Surface area bottom
	A_{total}	Surface area total
	$V_{nominal}$	Nominal voltage

	$\rho_{E,m}$	Specific energy
Remove T+E	E	Energy
	$\rho_{E,V}$	Energy density
	A_{side}	Surface area casing side
	A_{top}	Surface area top
	A_{bottom}	Surface area bottom
	A_{total}	Surface area total
	$V_{nominal}$	Nominal voltage
	$\rho_{E,m}$	Specific energy

Table. S4 Feature selection test design

Constructed features	Definition	Formula	Unit	Source term
V_{cell}	Cell volume	$\frac{E_{cell}}{\rho_{E,V}}$	cm^3	Energy, energy density
E_{store}	Stored energy	$OCV \times C \times SOC$	Wh	OCV, capacity, SOC
$E_{store,norm}$	Normalized stored energy	$\frac{OCV \times C \times SOC}{V_{cell}}$	$\frac{Wh}{cm^3}$	OCV, capacity, SOC, energy, energy density

Table. S5 Thermal features + constructed features (with redundancy)

Test Name	Features	Definition
TC1	m_{cell}	Pretest cell mass
	$m_{cm,p}$	Positive copper mesh mass
	$m_{cm,n}$	Negative copper mesh mass
	Q_{heater}	Heater power
	t_{heater}	Heater time
	V_{cell}	Cell volume
TC2	m_{cell}	Pretest cell mass
	$m_{cm,p}$	Positive copper mesh mass
	$m_{cm,n}$	Negative copper mesh mass
	Q_{heater}	Heater power
	t_{heater}	Heater time
	E_{store}	Stored energy
TC3	m_{cell}	Pretest cell mass
	$m_{cm,p}$	Positive copper mesh mass
	$m_{cm,n}$	Negative copper mesh mass
	Q_{heater}	Heater power
	t_{heater}	Heater time
	$E_{store,norm}$	Normalized stored energy

TC4	m_{cell}	Pretest cell mass
	$m_{cm,p}$	Positive copper mesh mass
	$m_{cm,n}$	Negative copper mesh mass
	Q_{heater}	Heater power
	t_{heater}	Heater time
	V_{cell}	Cell volume
	E_{store}	Stored energy

Table. S6 Electrochemical features + constructed features (with redundancy)

Test Name	Features	Definition
EC1	OCV	Pretest open-circuit voltage
	Q	Pretest capacity
	SOC	Pretest state of charge
	V_{cell}	Cell volume
EC2	OCV	Pretest open-circuit voltage
	Q	Pretest capacity
	SOC	Pretest state of charge
	E_{store}	Stored energy
EC3	OCV	Pretest open-circuit voltage
	Q	Pretest capacity
	SOC	Pretest state of charge
	$E_{store,norm}$	Normalized stored energy
EC4	OCV	Pretest open-circuit voltage
	Q	Pretest capacity
	SOC	Pretest state of charge
	V_{cell}	Cell volume
	E_{store}	Stored energy

Table. S7 Thermal features + constructed features (without redundancy)

Test Name	Features	Definition
TCR1	m_{cell}	Pretest cell mass
	$m_{cm,p}$	Positive copper mesh mass
	$m_{cm,n}$	Negative copper mesh mass
	Q_{heater}	Heater power
	t_{heater}	Heater time
	V_{cell}	Cell volume
	E_{store}	Stored energy
TCR2	m_{cell}	Pretest cell mass
	$m_{cm,p}$	Positive copper mesh mass
	$m_{cm,n}$	Negative copper mesh mass
	Q_{heater}	Heater power
	t_{heater}	Heater time
	$E_{store,norm}$	Normalized stored energy

Table. S8 Electrochemcial features + constructed features (without redundancy)

Test Name	Features	Definition
ECR1	OCV	Pretest open-circuit voltage
	Q	Pretest capacity
	SOC	Pretest state of charge
	V_{cell}	Cell volume
ECR2	V_{cell}	Cell volume
	E_{store}	Stored energy
ECR3	$E_{store,norm}$	Normalized stored energy

Table. S9 Feature construction

Test Name	Features	Definition
C1	V_{cell}	Cell volume
C2	E_{store}	Stored energy
C3	$E_{store,norm}$	Normalized stored energy
C4	V_{cell}	Cell volume
	E_{store}	Stored energy

Tabel. S10 Experiment cost with pretest features

Tasks	Function	Price ($)	Source
Cell price	Buy cells	2028	https://www.18650batterystore.com/search?type=product&q=LG
Buy device	Buy cycler	2,000	https://www.goldsupplier.com/provide/p159126923.html
Pretest measurement	Obtain pretest features	1720	consume time: 2h, PhD hourly salary: $20
Total		**5748**	

Table. S11 Experiment cost with post-test features

Tasks	Subtasks	Function	Price ($)	Source
Cell price	-	Buy cells	2028	https://www.18650batterystore.com/search?type=product&q=LG
Buy device		Buy cycler	2,000	https://www.goldsupplier.com/provide/p159126923.html
Pretest measurement	-	Obtain pretest features	1720	consume time: 2h, PhD hourly salary: $20
Buy device		Build FTRC	10,000	EMSLab
FTRC	setup - 2h		13520	consume time: 10h PhD hourly salary: $20
	exp - 0.5h			
	collect data + clean up - 2h			
	4 people			
Post-test	data process - 4h	27040	consume time: 4h PhD hourly salary: $20	
Total			**56,308**	

Table. S12 Hyperparameters range of used algorithms

Algorithm	Hyperparameter	Candidate values / range
ElasticNet	alpha	0.001, 0.01, 0.1, 1, 10
	l1_ratio	0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1.0
SVR	C	0.1, 1, 10
	gamma	scale, auto
	epsilon	0.01, 0.1, 0.5
	kernel	rbf, linear
LightGBM	learning_rate	0.01, 0.1
	num_leaves	31, 50
	max_depth	−1, 5, 10
XGBoost	learning_rate	0.01, 0.05, 0.1
	max_depth	3, 5, 7
	subsample	0.8, 1.0
	colsample_bytree	0.8, 1.0
KNN	n_neighbors	3, 5, 7
	weights	uniform, distance
MLP	hidden_layer_sizes	(10,), (20,), (20, 20)
	activation	relu, tanh
	alpha	0.001, 0.01
	learning_rate_init	0.001, 0.01

References

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